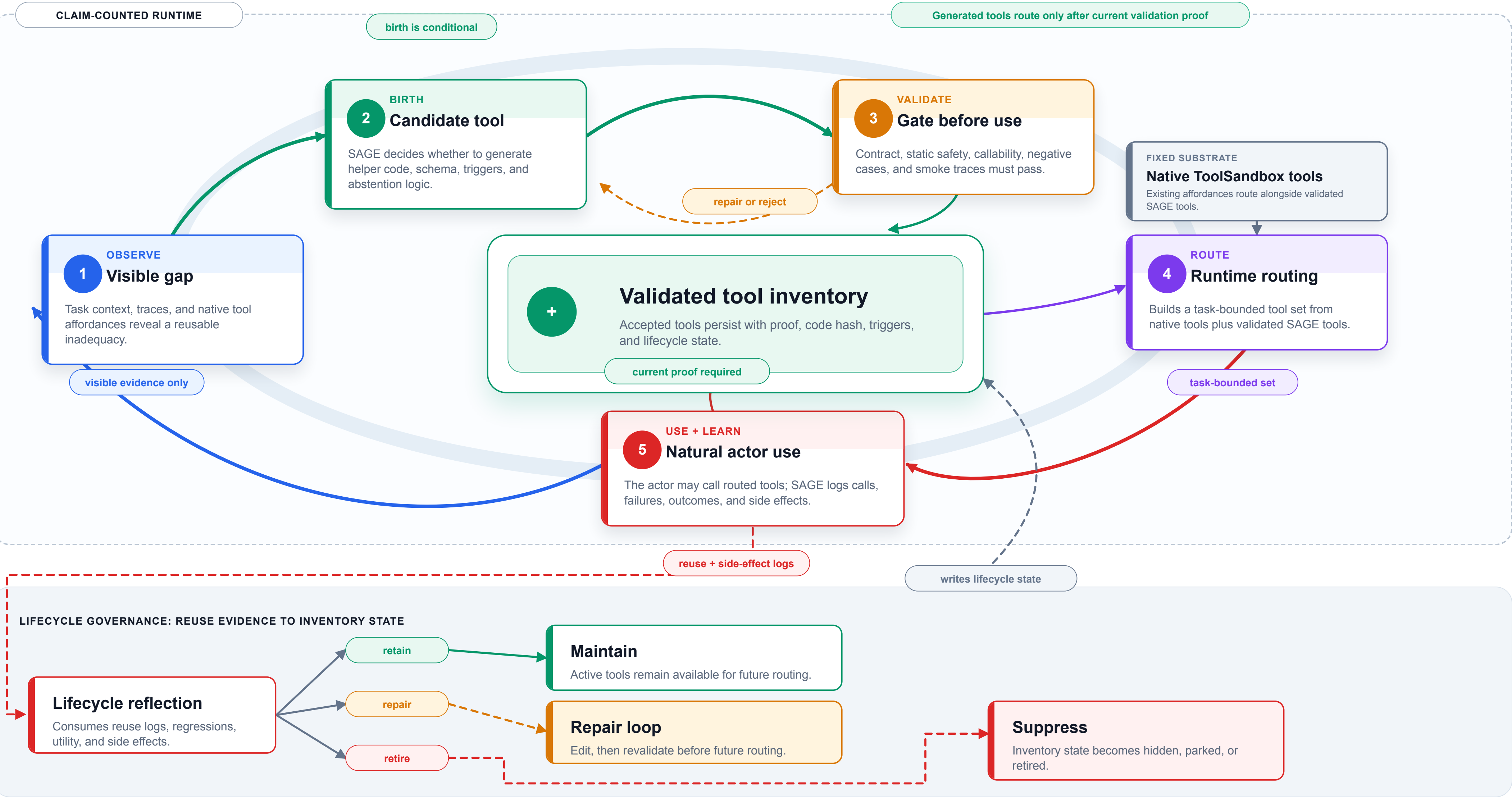


SAGE Generated-Tool Self-Evolution Loop

A governed flywheel: visible task context reveals an inadequacy, SAGE births and validates a generated tool, routes accepted tools for natural actor use, and updates lifecycle state from reuse evidence.



Paper boundary: generated tools use visible context, pass validation before routing, preserve native side-effect tools, and remain subject to lifecycle repair or retirement.